

Course and its outcomes

III Semester: Nuclear and Particle Physics (B010902T)

Unit I	Deuteron, n-n scattering, n-p scattering, p-p scattering, charge symmetry of nuclear forces.
Unit II	Shell Model, Extreme Single particle picture and angular momentum, magnetic moment, quadrupole moment of nuclei, Nuclear Isomerism, Collective model (qualitative discussion)
Unit III	Compound Nucleus, Breit-Wigner Formula, Direct Interaction, Heavy Ion Reactions, Relativistic Kinematics
Unit IV	Fundamental types of Interactions, General Classifications of Elementary Particles, Isospin, Strangeness, Conservation Laws, Symmetries (C, CP, CPT), SU(3) and quark model
Unit V	Alpha, beta and gamma decay

Reading List:

1. Nuclear Physics, 2nd edition, V. Devnathan (Narosa, 2011)
2. Introduction to Elementary Particles, 2nd Edition, D. J. Griffiths (Wiley-VCH, 2008)
3. Introductory Nuclear Physics, 3rd edition (An Indian Adaptation), Kenneth S. Krane, (Wiley India Pvt. Ltd., 2022)
4. Introductory Nuclear Physics, 2nd edition, Samuel S.M. Wong, (Wiley India Pvt. Ltd., 2013)
5. Nuclear Physics, D.C. Tayal (Himalaya Publishing House, 2009)

Learning Outcomes

Knowledge:

The students

- Will have understanding of n-n, n-p and p-p scattering and charge symmetry of nuclear forces.
- Will be able to explain reasons behind ground state properties of nucleus such as spin-parity assignment, angular moment and magnetic moment with the help of shell model.
- Will have understanding of conservation laws associated with reactions and decays of elementary particles.
- Will have fundamental understanding of nuclear decays and associated selection rules.

Skills:

The students

- Will be able to determine spin-parity assignments and magnetic moment of nuclei in ground state with the help of shell model and collective model.
- Will be able to analyze kinematics formulation associated with relativistic heavy ion reactions.
- Will be able to predict if an elementary particle reaction is allowed and to recognize the type of the interaction associated with it.